

Realised Volatility Forecasting: A Liquidity-Regime Approach to Model Selection

DATA3888 Group 8

Executive Summary

Realised volatility forecasting estimates how much an asset price is likely to move over a short future interval. This matters for market making, risk control, and trading decision support because a trader must decide whether the next few minutes are likely to be stable or turbulent. In high-frequency order-book data, volatility is not only a price-history problem. It is also a market microstructure problem: a stock with a tight bid-ask spread and deep activity behaves differently from a stock with a wide spread and thin activity.

Our project therefore reframes the modelling question. Instead of asking which forecasting model is best for every stock, we ask which model is appropriate under the current liquidity regime. We use bid-ask spread (BAS), activity proxies, and realised volatility (RV) to classify stocks and sessions into liquid, mixed, and illiquid conditions. The latest regime configuration identifies 39 liquid stocks, 30 mixed stocks, and 43 illiquid stocks in the 112-stock universe, and selects 20 stocks from each regime for a balanced 60-stock demonstration split.

The main empirical result from the current model outputs is that LightGBM is the strongest predictive benchmark. On the 40-stock liquid/illiquid model outputs currently present in the repository, LightGBM has the lowest median QLIKE and MSE for all overlapping liquid and illiquid stocks. However, the model-selection decision is not only an accuracy race. EGARCH-X is more interpretable for liquid stocks because it models conditional volatility, asymmetry, and the spread channel directly. HAR-RV and WLS provide stable fallback models for illiquid settings where GARCH-family estimation can be fragile. The deployed trading-assistant idea therefore uses a liquidity-aware router: recommend interpretable statistical models when stability and explanation are required, but show LightGBM as the accuracy benchmark and possible override.

Background and Data

The analysis uses an aggregated Optiver order-book dataset with stock identifiers, time identifiers, 30-second time buckets, weighted average price (WAP), bid-ask spread, order activity proxies, and realised volatility. Each time ID represents a 10-minute trading window split into 20 buckets. The shared modelling design uses the first 16 buckets, or 8 minutes, for training information and the final 4 buckets, or 2 minutes, for validation. This matches the product setting: a trader observes recent market conditions and needs a near-term volatility forecast.

The original workflow proposal established the baseline sequence: raw order-book data to WAP, log returns, 30-second realised volatility, an 80/20 time split, model fitting, and evaluation by QLIKE and MSE. Week 9 then shifted the project from a single-model comparison to a regime-aware system after GARCH-family models showed instability in thin order books. Week 11 added a three-regime requirement: liquid, mixed, and illiquid stocks should be separated rather than forced into a binary classification.

Liquidity is measured by two complementary signals. BAS measures trading cost and order-book tightness. Activity is proxied through inverse spread and WAP-based quantities; higher activity and lower spread indicate deeper liquidity. In the latest regime file, a stock is liquid if at least 40% of its buckets satisfy the liquid condition, illiquid if at least 40% satisfy the illiquid condition, and mixed

otherwise. This prevents a stock with occasional thin sessions from being treated as permanently illiquid.

Table 1: Three-regime summary from the pulled stock-selection file. The full universe counts are 39 liquid, 30 mixed, and 43 illiquid stocks; the table reports the balanced 60-stock demonstration subset used for the new shared design.

Regime	Demo stocks	Full-universe share	Median BAS	Median log activity	Mean regime share
Liquid	20	34.8%	0.000259	8.256	70.5% liquid
Mixed	20	26.8%	0.000417	7.781	55.7% mixed
Illiquid	20	38.4%	0.000782	7.153	65.9% illiquid

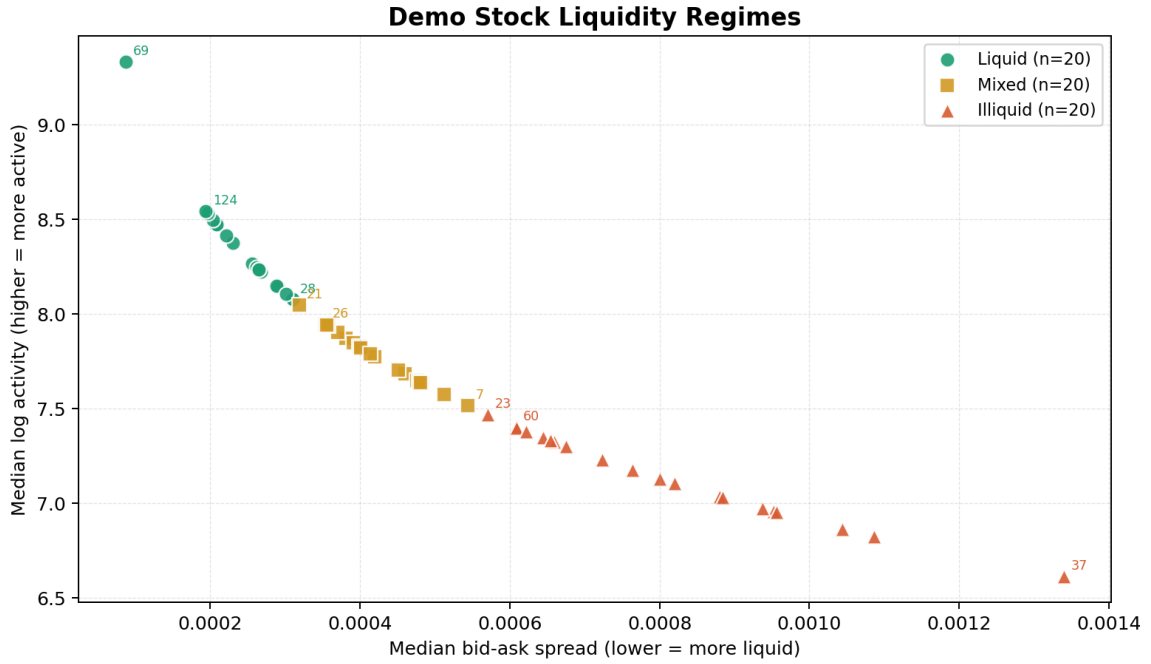


Figure 1: Liquidity regimes in the balanced demonstration subset. Liquid stocks cluster at lower BAS and higher activity; illiquid stocks occupy the wide-spread, lower-activity region; mixed stocks sit between these extremes.

Methods

Validation Design

The final holdout structure is an 80/20 chronological bucket split: buckets 1–16 are used as training information and buckets 17–20 are used as validation. We prefer this to ordinary k -fold cross-validation because a trading system never has future buckets available when making a forecast. Random folds would mix future and past microstructure states, leading to look-ahead leakage and overly optimistic performance.

We also considered rolling-window validation. It is attractive when the system dynamics change over long horizons, but here each time ID contains only 10 minutes. Rolling many near-identical windows inside a short session can repeatedly evaluate the same microstructure noise and produce highly correlated test sets. The regime design addresses a different source of non-stationarity: instead of asking the model to adapt mid-window, we first identify the liquidity scenario and then evaluate models within that scenario. Time-ordered cross-validation is still useful as a diagnostic inside the training period, but the main model comparison uses the fixed 8-minute to 2-minute forecast task.

Diagnostics, Metrics, and Selection Criteria

We used graphical, likelihood-based, and loss-based evaluation. ACF and PACF plots were used as time-series diagnostics before fitting the GARCH-family models. The ACF at lag k measures the correlation between the series and its k -lagged value; persistent positive autocorrelation suggests that volatility information remains useful across buckets. The PACF measures the incremental lag- k relationship after controlling for shorter lags, helping decide whether a low-order autoregressive mean component is sufficient. These diagnostics motivated a simple ARMA(1,1) mean structure rather than a high-order model on only 16 training buckets.

For GARCH-family order selection, we used AIC and BIC. Both are likelihood criteria that reward fit but penalise extra parameters:

$$\text{AIC} = 2k - 2 \log L, \quad \text{BIC} = k \log(n) - 2 \log L,$$

where k is the number of fitted parameters, n is the sample size, and L is the maximised likelihood. Lower values are preferred. BIC was the final selector because it penalises complexity more strongly than AIC, which is important when each time-ID fit has only 16 training buckets and over-parameterised models can converge unreliably.

The main forecast losses are MSE and QLIKE. MSE is the average squared forecast error and is easy to interpret in the units of volatility. QLIKE is a volatility-specific loss that heavily penalises under-prediction of variance; in variance form it is $\log(\hat{\sigma}_t^2) + \sigma_t^2 / \hat{\sigma}_t^2$ up to constants. Lower values are better. We also evaluate convergence, parameter sanity checks, and blow-up rates because a model with good average error is not deployable if a small number of predictions become numerically unstable.

Model Families

GARCH baseline. The baseline is an ARMA(1,1)-GARCH model fitted to realised-volatility series within each stock and time ID. GARCH models conditional variance through past squared shocks and past variance, so they are designed for volatility clustering. The current code tunes GARCH orders $(p, q) \in \{1, 2\}^2$ on 200 sampled time IDs and selects by mean BIC. The selected order is GARCH(1,1), with mean BIC -29.587 .

EGARCH-X refinement. EGARCH models log conditional variance, allowing asymmetric responses through a leverage parameter. Our EGARCH-X adds a spread term:

$$\log(\sigma_t^2) = \omega + \alpha(|z_{t-1}| - E|z|) + \gamma z_{t-1} + \beta \log(\sigma_{t-1}^2) + \delta \log(\text{BAS}_{t-1}^2).$$

Here γ measures asymmetry: negative shocks may raise volatility differently from positive shocks. The δ term measures the liquidity channel: wider spreads can directly raise predicted conditional variance. EGARCH-X is therefore economically meaningful for liquid stocks, where spread data are reliable enough to estimate this channel. The current global BIC tuning selects EGARCH-X(2,1,2). The implementation also rejects degenerate fits with extreme γ , δ , or near-unit persistence, because short 16-bucket training samples can make optimisers report convergence at unstable boundaries.

HAR-RV and WLS. HAR-RV predicts future RV using short-, medium-, and full-window realised-volatility components. In our 30-second bucket setting, these are adapted as the last bucket, the mean of the last five buckets, and the mean over the 16-bucket training window. The checked-in `rosa.py` implementation fits HAR-RV by OLS per time ID and floors invalid predictions to avoid QLIKE blow-ups. Jamie’s WLS pipeline is a separate regression baseline using engineered microstructure features. It tunes an exponential recency weight parameter α over a grid and fits weights α^{16-b} for bucket b , so recent buckets receive more influence. This is justified because volatility from the most recent minute is usually more informative about the next two minutes than volatility near the start of the 10-minute window.

LightGBM benchmark. LightGBM is used as a flexible non-linear benchmark. It uses lagged and summary RV features, spread summaries, spread trends, WAP summaries, and stock-level global features. Important engineered features include `rv_mean`, `rv_last`, `rv_last3_mean`, `spread_last`,

and `spread_trend`. The current implementation trains separate models for liquid and illiquid regimes in the 40-stock pipeline, with 500 trees, learning rate 0.05, 63 leaves, subsampling and column sampling of 0.8, and L1/L2 regularisation of 0.1. These settings are conservative controls against overfitting rather than a claim of exhaustive hyperparameter optimisation.

Results

The liquidity regimes are meaningful. Figure 1 shows a smooth movement from liquid stocks with low spreads and high activity to illiquid stocks with wide spreads and lower activity. This supports the Week 11 decision to include a mixed regime rather than routing all borderline stocks into liquid or illiquid categories.

Table 2: Current model-output summary from repository CSVs. These metrics are from the current 40-stock liquid/illiquid outputs, not yet the newly pulled 60-stock three-regime rerun. Lower QLIKE and MSE are better.

Regime	Model	Stocks	Median QLIKE	Median MSE
Liquid	LightGBM	20	-14.528	3.57×10^{-9}
Liquid	GARCH	20	-12.678	2.09×10^{-6}
Liquid	EGARCH-X	20	-12.974	1.20×10^{-6}
Illiquid	LightGBM	20	-12.884	3.95×10^{-8}
Illiquid	GARCH	20	-9.519	5.13×10^{-5}
Illiquid	HAR-RV	20	-5.492	1.94×10^{-7}

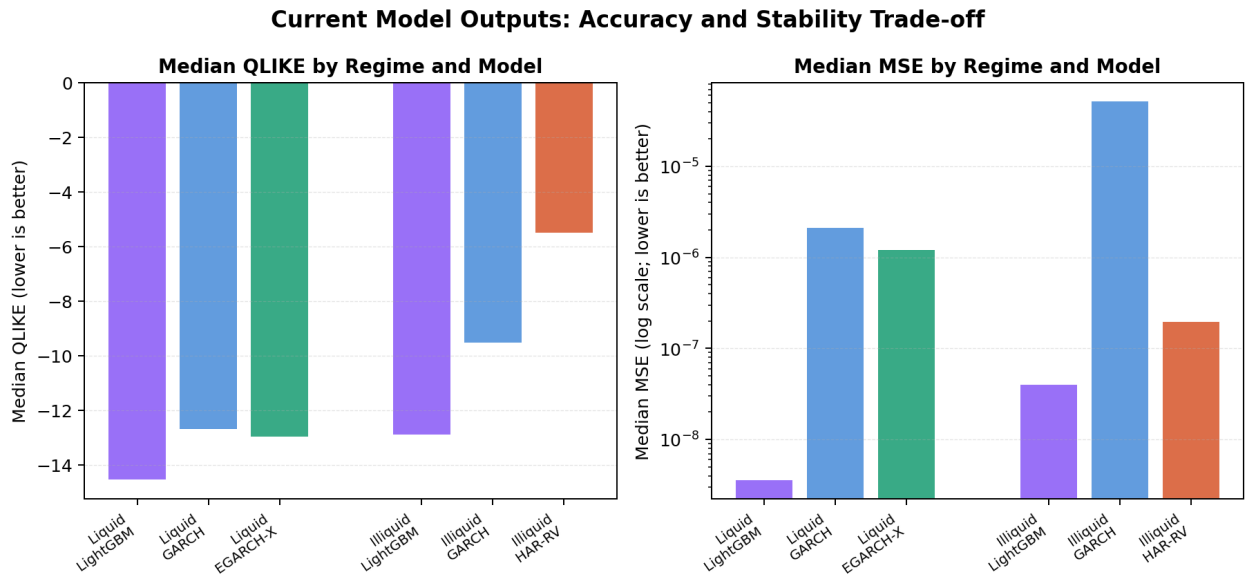


Figure 2: Median performance from the current output CSVs. LightGBM is the strongest predictive benchmark in both regimes. GARCH has better QLIKE than HAR-RV in the illiquid output, but much worse MSE and flagged blow-ups, so it is not automatically the safer illiquid recommendation.

For liquid stocks, EGARCH-X improves on GARCH in median QLIKE and MSE, which supports adding the spread channel and asymmetry term. However, LightGBM is better than both statistical models in the current outputs. This should be stated directly in the presentation: if the goal is pure predictive accuracy, LightGBM currently wins; if the goal is interpretable volatility mechanics, EGARCH-X remains useful because its parameters explain volatility persistence, leverage, and the BAS effect.

For illiquid stocks, GARCH appears better than HAR-RV by the current QLIKE values, but this is not the whole story. HAR-RV has much smaller median MSE, is easier to fit, and avoids spread-driven instability. GARCH also flags approximately 1.0% of illiquid forecasts as top-tail QLIKE blow-ups. The practical recommendation is therefore conditional: use LightGBM when accuracy and a trained benchmark are available; use HAR-RV or WLS as the stable fallback when interpretability and failure resistance are more important; avoid relying on GARCH alone in thin order books.

This leads to the following final model-selection rule for the current evidence. In liquid regimes, EGARCH-X is the preferred interpretable statistical model because it uses BAS as an external volatility driver and passes stability checks more often than in thin markets; LightGBM is shown beside it as the accuracy benchmark. In mixed regimes, the safest current statement is that LightGBM should be the benchmark until all statistical models are rerun on the new 60-stock split, because mixed stocks are precisely where linear regime rules are least reliable. In illiquid regimes, HAR-RV/WLS are the stability-first recommendations, while LightGBM is the accuracy-first option. If the selected regime model performs badly for a stock, the dashboard should detect this through high recent QLIKE/MSE, convergence failures, or blow-up flags, then fall back to the next available model with lower validation loss and a valid stability profile.

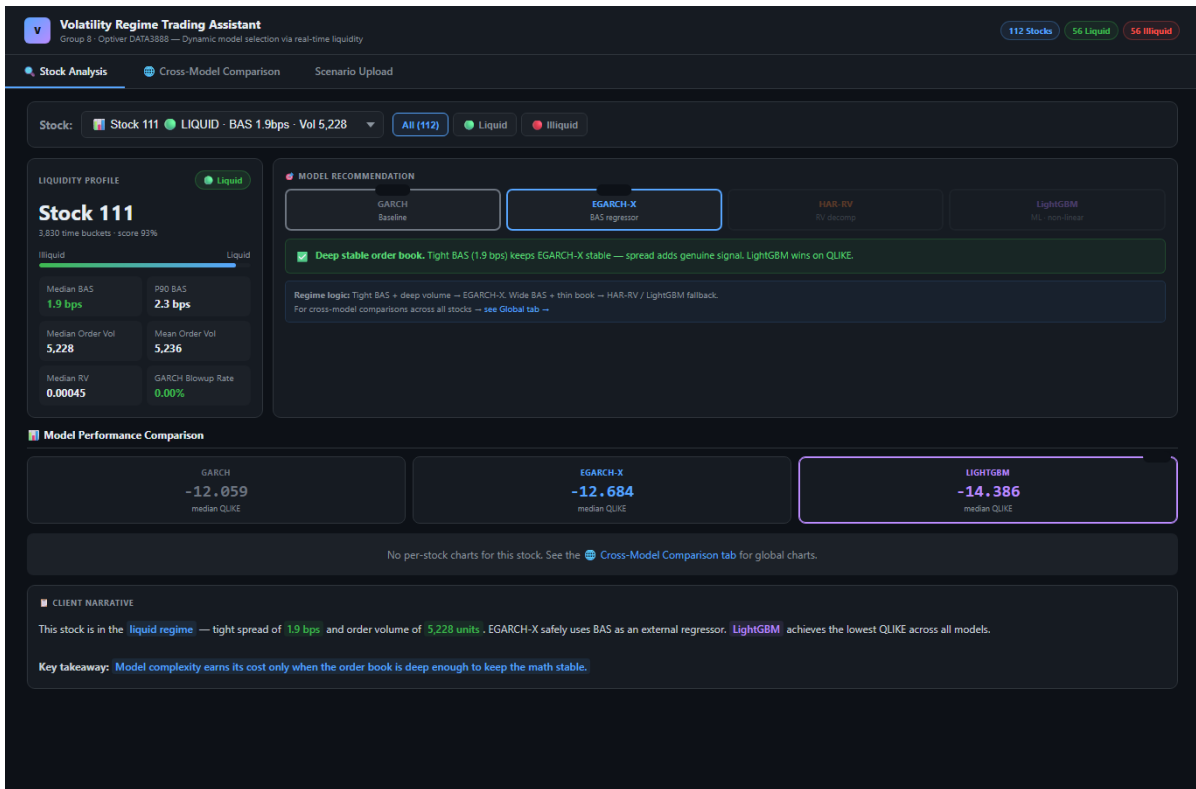


Figure 3: Prototype trading assistant. The current screenshot is from the earlier binary liquid/illiquid dashboard; the next integration step is to add the mixed-regime counts and routing. The dashboard reads stock-level liquidity summaries, displays model diagnostics, and shows both the regime-based statistical recommendation and the LightGBM benchmark.

Deployment

The product is a lightweight HTML trading assistant rather than an automated trading system. A user selects a stock and sees its liquidity profile, BAS, order activity, GARCH blow-up rate, model diagnostics, and recommended forecast model. The router has three steps: compute liquidity features, classify the stock or session into a regime, then display the model recommendation and benchmark comparison. This makes the modelling decision auditable: the user can see why a model

was selected and whether an alternative has lower historical loss.

Discussion and Conclusion

The main scientific insight is that volatility forecasting is regime-dependent. High-frequency data violate several textbook assumptions: normal residuals, stable stationarity, symmetric effects, and equal relevance of all past observations. Our modelling choices respond to these violations. BIC tuning prevents unnecessarily complex GARCH orders on very short series. EGARCH-X captures asymmetry and spread effects. WLS gives more weight to recent buckets. LightGBM captures non-linear interactions and provides a strong accuracy benchmark.

There are also important limitations. First, some current model outputs were produced on the earlier 40-stock binary liquid/illiquid pipeline, while the latest pulled design introduces 60 stocks across liquid, mixed, and illiquid regimes. The final claim about mixed stocks should wait until the models are rerun on that shared three-regime split. Second, the QLIKE implementation should be harmonised across all scripts before final submission: GARCH and LightGBM currently use a squared-volatility form, while HAR-RV/WLS code uses the variance form. This does not remove the observed MSE and stability patterns, but it affects direct QLIKE ranking across model families. Third, LightGBM's strong performance should be balanced against interpretability and validation design. If the production goal is transparent risk explanation, a black-box benchmark may not be the only final answer.

In conclusion, our project builds a liquidity-aware volatility forecasting system. The best current empirical predictor is LightGBM, but the most useful product is a router that communicates the trade-off between accuracy, interpretability, and stability. Liquid stocks can support richer models such as EGARCH-X; illiquid stocks need stable fallbacks such as HAR-RV or WLS; mixed stocks should be treated as transition cases where LightGBM and interpretable models are compared after the shared 60-stock rerun.

Student Contributions

Jamie developed the liquidity-regime classification, WLS feature engineering, and the updated liquid/mixed/illiquid stock-selection design. Jisu implemented the GARCH and EGARCH-X modelling, BIC tuning, diagnostics, and blow-up tracking. Rosa implemented HAR-RV modelling and evaluation. Gia implemented the LightGBM benchmark and feature-importance outputs. Skylar built the trading-assistant dashboard. Sabrina coordinated the explanatory narrative and presentation/report communication. All members should confirm this paragraph before submission.

Reproducibility and Appendix

The main analysis artifacts are reproducible from the repository scripts and output files. The key files are:

- `generate_splits.py`: shared 60-stock three-regime train/test split.
- `eda_and_wls_jamie.py`: three-regime EDA, feature selection, and WLS pipeline.
- `arma_jisu.py`: GARCH and EGARCH-X fitting, BIC tuning, and diagnostics.
- `rosa.py`: HAR-RV fitting and evaluation.
- `lgbm.py`: LightGBM benchmark and feature importance.
- `pipeline.py` and `trading_assistant.html`: model-router outputs and dashboard prototype.

References

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